

## STANDARD OPERATING PROCEDURE FOR BOILER

### Boiler Specifications:

**Type of Boiler:** Water tube boiler (Horizontal)

**Manufacture:** Thermax

**Capacity:** 3MT/ Hr.

**Max working pressure:** 11.25 kg/cm<sup>2</sup>, but present operating pressure is 5 kg/cm<sup>2</sup>

- Relevant personal protective equipment to be used by operator
- Boiler should operate first class certificate Operator
- Ensure the feed water PH value is around 8.5 to 9.5.
- Ensure day soft water tank is filled to an adequate level.
- The water flow in tubes should be operated by the help of globe valve.
- Turn the main switch of Boiler to “ON” position on the boiler panel and press control “ON” green button, the light will glow
- Open the water inlet valve.
- Ensure availability of required coal in bunker.
- From the raw material yard, the coal is shifted through the conveyor belt to the bunker as required.
- Then the coal from the bunker is supplied to furnace using coal feeder conveyor.
- Coal is ignited in the furnace.
- Forced fan is used for circulation of heat produced by the coal to the header.
- For removing of exhaust gases from the furnace, the gases are to be blown out through chimney using induced draught.
- The exhaust gases which are leaving the boiler at such high temperature is made to pass through the Economizer in order to provide the required sensible heat to the water by increasing its temperature, it will reduce the heat load on the boiler.
- Pressure gauges and valves should check at regular intervals of time by the operator, Steam pressure 4.5Kg/cm<sup>2</sup> and all other parameters are to recorded
- Steam is sent to the pickling process by the help of Pressure reducing valve.
- Note down the all parameter for every hour in logbook.